

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Embedded Systems
COURSE CODE: ECA1422

COURSE OUTCOME COVERED

CO1: Analyze and design complex and networked embedded systems using formal design methodologies, embedded architectures, CAN (BLS)

Total Marks: 20

Sl. No	QUESTION BANK	BLOOMS LEVEL
16	create a concept map connecting embedded product reliability concepts - including failure rate, MTTF, reliability models, bathtub curve, FMEA, fault tree analysis, and redundancy strategies - and establish how these reliability engineering tools influence design decision across the embedded product life	

Code of Conduct:

KEERTHIPATI HEMA SREE (192411116) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

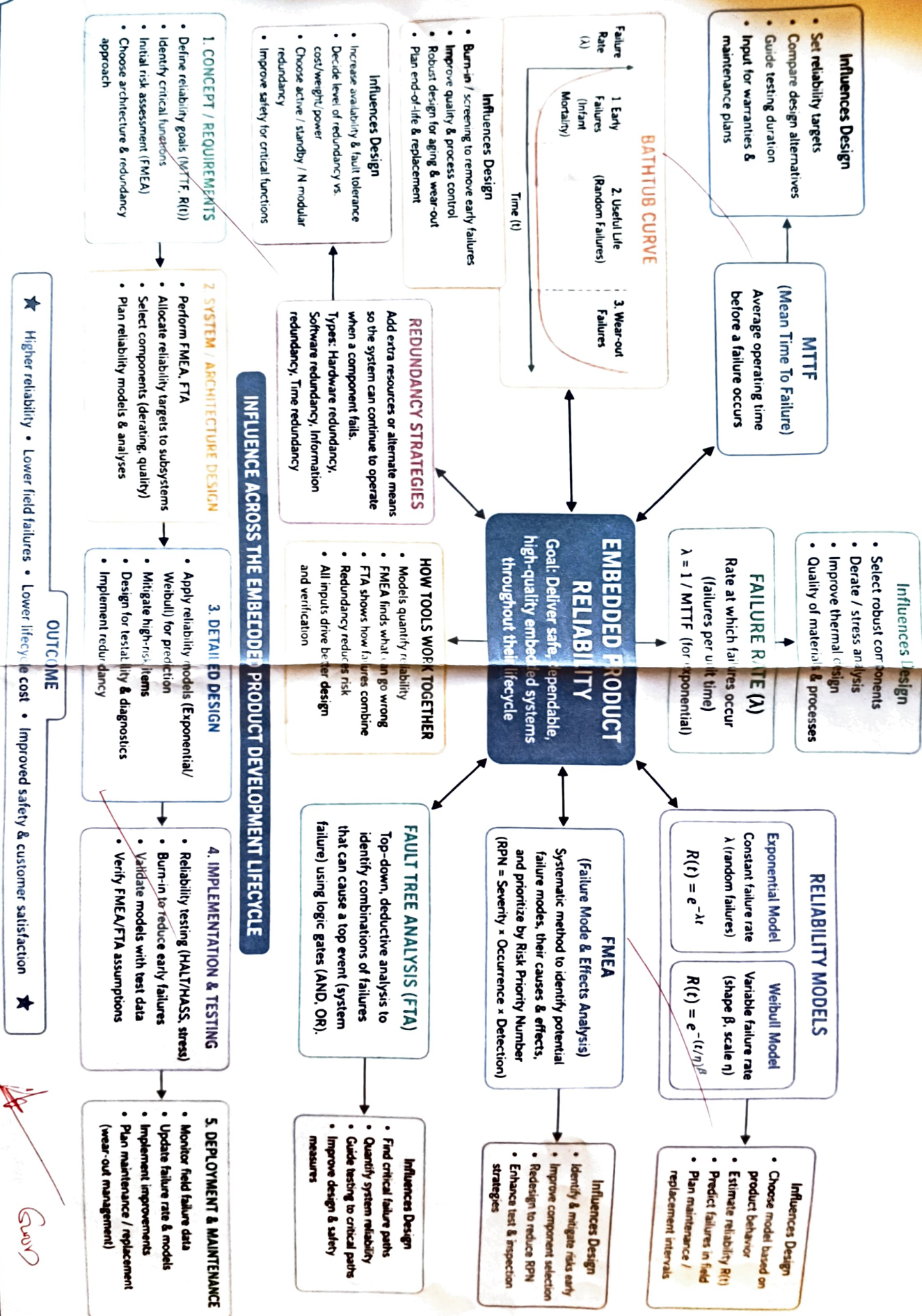
Evaluation Rubric (20 Marks per Question – Concept Map)

Criteria	Weightage	Level 4 (Exemplary) (5 marks)	Level 3 (Proficient) (4 marks)	Level 2 (Developing) (3 marks)	Level 1 (Beginning) (1-2 marks)
Concept Identification	25 %	25			
Linkages	25 %	25			
Organization	20 %	20			
Integration of Literature	20 %	20			
Clarity	10%	10			

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EMBEDDED PRODUCT RELIABILITY

Concept Map: Tools, Models & Their Influence Across the Development Lifecycle



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